

Vaibhav Balloli

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Research Focus

AI researcher focusing on reliability, safety, experience, and usability of AI systems in human-AI in social good applications. My PhD research focuses on improving conversational (human-LLM) and agentic capabilities for maternal health.

Research Topics: LLMs Evals, Governance, Safety, and Deployment}, human-AI interaction

Education

University of Michigan, Ann Arbor

2023 - Present

Ph.D. Candidate, M.S. in Computer Science and Engineering

Advisor: [Prof. Elizabeth Bondi-Kelly](#)

Courses: Reinforcement Learning Theory, Causality and ML, NLP, HCI, AI, Advanced Data Mining

BITS Pilani, Hyderabad Campus

2016 - 2020

B.E in Electronics and Communication Engineering

Research Experience

Microsoft Research Redmond, AI Frontiers

May 2025 - Aug 2025

Research Intern

Advisors: [Hussein Mozannar](#), [Adam Fourney](#), [Gagan Bansal](#), [Saleema Amershi](#) & [Eric Horvitz](#)

Topic: Multi-turn Reinforcement Learning with Large Language Models Agents, Agent evaluations

Microsoft Research India

June 2022 - June 2023

Research Fellow

Advisors: [Dr. Akshay Nambi](#) & [Tanuja Ganu](#)

Topics: Reinforcement Learning, Large Language Models.

Projects and Press: [Vasudha](#), [VeLLM](#): [VeLLM Press](#) - [Satya Nadella](#), [Times of India](#)

Impact: VeLLM is currently being deployed in schools across India

Microsoft Research India

June 2021 - June 2022

Societal impact through Cloud and AI Fellow

Advisors: [Dr. Akshay Nambi](#), [Tanuja Ganu](#) & [Dr. Venkat Padmanabhan](#)

Topics: Computer Vision, Bandits, Localization

Projects and Press: [HAMS](#): [HAMS Press](#) - [Punjab News Express](#)

Impact: HAMS is successfully deployed across India - [Dashboard](#) (300,000 users)

Working papers

1. **To Ask or Not To Ask: Evaluating Information Seeking, Execution, and Abstention Capabilities in Long-Context LLM Agents**
[Vaibhav Balloli](#), [Elizabeth Bondi-Kelly](#)
2. **Towards Improving Interactivity in Multi-Turn Human-Agent Systems**
[Vaibhav Balloli](#), [Hussein Mozannar](#), [Adam Fourney](#), [Gagan Bansal](#), [Saleema Amershi](#), [Eric Horvitz](#)
Topics: *Large Language Models, Agents, GRPO*

Selected Publications

6. **RELIANCE: Curating and Evaluating Reproductive Health Information on Social Media**
Vaibhav Balloli, Laura Peyton Ellis, Vishala Mishra, Alice Chi, Alex Friedman Peahl, Elizabeth Bondi-Kelly
Topics: Large Language Models, Reproductive Health, Multimodal eval.
KDD Datasets and Benchmarks, 2026
5. **PATHFinder Agent for Tailored Prenatal Care**
Vaibhav Balloli, Carissa Samuel, Samia Abdelnabi, Alex Peahl, Elizabeth Bondi-Kelly.
Topics: Large Language Models Agents, Long-context optimization, Deployment
Interactive Health, 2026
4. **“Where is this coming from?” Uncovering Trustworthiness Ideals in AI-powered Peripartum Information Seeking**
Vaibhav Balloli, Julia Erickson, Xinyi Li, Erin van Liemt, Alex Friedman Peahl, Elizabeth Bondi-Kelly
Topics: Socio-technical governance, human-AI Interaction
ACM FAccT, 2026
3. **Bridging the Gap: Dynamic Learning Strategies for Improving Multilingual Performance in LLMs**
Somnath Kumar*, **Vaibhav Balloli***, Mercy Ranjit, Kabir Ahuja, Tanuja Ganu, Sunayana Sitaram, Kalika Bali, Akshay Nambi
Topics: Large Language Models, Multilingual Evaluation, Human Feedback, Representation Learning
Oral presentation, COLING 2025
2. **Are They the Same Picture? Adapting Concept Bottleneck Models for Human-AI Collaboration in Image Retrieval**
Vaibhav Balloli, Sara Beery, Elizabeth Bondi-Kelly
Topics: human-AI Collaboration, Representation Learning, Interpretability, Wildlife Conservation
IJCAI'24 Human-Centered AI Track, CV4Animals Workshop@CVPR'24 | [Code](#) | [Project](#)
[<5% Acceptance Rate]
1. **Chanakya: Learning Tradeoffs for Adaptive Streaming Perception via Contextual Bandits**
Anurag Ghosh, **Vaibhav Balloli**, Akshay Nambi, Aditya Singh, Tanuja Ganu
Topics: Object detection and tracking, Computer Vision, Multi-arm bandits
NeurIPS'23 | [Talk](#) | [Code](#)

Select Awards and Honors

- FAccT travel award 2026
- Exceptional Preliminary exam (advancement to candidacy exam) 2025
- Presented poster at Society-centered AI workshop at Google, Mountain View 2025
- 2nd Place in the Google x MHacks Hackathon - \$1500 [Press: Mentioned by labs.google](#) 2024
- [Press: VeLLM](#) mentioned by [Satya Nadella](#) and featured in [Times Of India](#), [FirstPost](#) 2024
- Selected for the [Harvard/MIT - HAIST/MAIA Intro Fellowship](#) on AI Safety 2023
- [Press: HAMS Automated License Testing](#) featured in [Punjab News Express](#) 2022
- [Winner of Microsoft Global Hackathon, 2021\(Future Of Edge Computing Track\)](#) 2021
- Co-founded **SmartCampus** which successfully managed **₹25 million** in transactions 2020

Service

- Conference Reviewer: NeurIPS'26, KDD'26, NeurIPS'25, IJCAI'25, ICML'25, AAMAS'25, ICLR'25 and NeurIPS'24.
- Workshop Reviewer: CV4Animals@CVPR'24, CompSust@ NeurIPS'23, EvalEval@ ACL'26